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A universal cooperative collision avoidance system for autonomous vehicles using horizontal manoeuvres

Mechanical Project 488

Project Proposal

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2026

forward together
sonke siya phambili
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Name: ADRIAN STARU Student no: 25855670

Date: 2026/03/02

Executive Summary

Title of Project
Design of a universal cooperative collision avoidance system for autonomous vehicles using horizontal manoeuvres only.
Objectives
Model a universal cooperative collision avoidance system and demonstrate it with wheel-based robotic vehicles at scale to show its feasibility.
What is current practice and what are its limitations?
Currently, there are proprietary collision avoidance systems used by a variety of industries and companies, that means that small autonomous applications like consumer drones do not contain this technology. Also, due to their proprietary nature, most collision avoidance systems do not account for other vehicles routes, only addressing the issue as it becomes imminent.
What is new in this project?
This project will aim to create a universal system that can be used across a variety of autonomous vehicles, not designed for one specific case. It will also be cooperative, allowing the vehicles to work together to decide on the most efficient path for all.
If the project is successful, how will it make a difference?
If successful, this project will allow cooperative collision avoidance to be implemented in many more applications, and at any scale, lowering the barrier to entry for the use of this technology.
What are the risks to the project being a success? Why is it expected to be successful?
A model that meets the performance criteria cannot be created. If found the model may not be demonstrable at small-scale. Time and budgetary risks are also present. All risks are mitigated by planning and making use of all available resources.
What contributions have/will other students made/make?
Previous students have considered cooperative collision avoidance in single applications. Their methods will be considered in the process of this project's completion.
Which aspects of the project will carry on after completion and why?
This project will prove that it is technically feasible, however it must be decided if it is economically feasible to continue and develop the product to deployable scale.
What arrangements have been/will be made to expedite continuation?
The chosen model/method and performance data will be provided. The project will be documented in sufficient detail that someone can scale it up to a deployable state.

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Introduction

Most vehicles operate in a world in which autonomous and non-autonomously controlled vehicles must inhabit the same space. For example, both traditional cars and newer cars with partial-self driving must use the same roads. This has led to complex collision avoidance systems that only react to the world right in front of them. This means that vehicles can only respond reactively to their surroundings and not proactively.

Reacting proactively will make these systems far more effective. If the vehicles involved in each environment can work together to determine the least intrusive paths to their destinations, the need for reactive collision avoidance can be negated. This does require all vehicles involved to be able to communicate with each other but there are scenarios where that is increasingly common, consider using fixed-wing mapping drones all flying over the same area, delivery drones servicing the same area from a distribution hub or carrier robots moving around on a factory floor. Since there are a variety of these scenarios, a universal cooperative collision avoidance system would make it easier for this technology to be effectively implemented in a system.

The project proposed is the creation and demonstration of a universal cooperative collision avoidance system for autonomous vehicles.

This project, which Mr Adrian Stark is doing as part of a Mechatronic Project 488, stems from a proposal by Prof JAA Engelbrecht. It plans to build on work done by previous master's students (Thando Mtimkulu and Merrick Hughes) who each investigated this technology for a specific use case.

This document explains the project's objectives, motivation and planning. The steps that are planned in this process, as well as the expected costs and time scales for the study are also outlined.

Project Aim and Objectives

As mentioned, the project's aim is to create a cooperative collision avoidance system for autonomous vehicles.

The system will make use of the vehicles location, direction, velocity and desired path to determine a new path for each vehicle to its destination with the smallest impact to all vehicles. The system will consider the vehicles minimum velocities and maximum turning rates.

It is important that this project will focus on scenarios in which the vehicles involved can slow down but cannot stop rapidly in their path. This is to ensure that the system can be

used in applications like fixed wing aircraft or vessels. The project is limited to horizontal manoeuvres so the traditional z-dimension cannot be used as a collision avoidance tactic.

The objectives of this project are therefore:

1) Create and test 3 rudimentary collision avoidance models.
2) Create a multibody simulation of the most feasible model accounting for physical phenomena.
3) Build a demonstration of the system using scale robots.

Planned Activities

1. Perform a literature and market review

Study the literature on cooperative collision avoidance systems paying attention to systems that deal with vehicles that cannot stop suddenly e.g. aeroplanes and boats.

2. Identify compatible applications

Identify the applications where this system is likely to be used to give greater context to the project and set performance criteria. This will provide understanding into the required operating range for the system. It will be used to set parameters like minimum speeds and maximum turn rates, so that they are applicable to the desired applications.

3. Compile design and performance criteria

Determine the requirements of the system in the chosen applications. Compile a list of evaluation criteria that can be used to measure the models' performance.

4. Select feasible models

Determine the 3 most feasible models to meet the design and performance requirements set out. These models may be solutions currently being employed or hybrid systems that make use of a combination of approaches.

5. Simulate feasible models to determine performance

Make use of software simulation tools, MATLAB and SIMULINK, to create rudimentary models to model their performance. These models will make general assumptions about mass, friction and other physical properties which limit their accuracy.

6. Select the highest performing model

Using the performance and evaluation criteria chosen previously, decide on a model that will most feasibly meet design and performance criteria.

7. Design Review

Consult with supervisor about the model chosen to move forward with, once consensus has been reached, the project can continue.

8. Create a multi-body performance simulation

Make use of software like Simscape Multibody to create a detailed simulation of the system to improve the accuracy and usability of the model. This model will account for physical phenomena such as slip, inertia and mass distribution.

9. Establish demonstration requirements and limitations

Understand what the requirements are to create the demonstration. Component and part selection to create the demonstration. This will include selecting a tracking technology that will be vital to meeting budgetary constraints.

10. Create a scale demonstration of the best model

Create a demonstration that can be used to visualize the system. This is likely to include scale wheel-based robots in a confined area, executing the selected collision avoidance system.

11. Validate simulation with demonstration

Validate the simulation's performance data with that of the demonstration. This is critical in determining the feasibility of the system and whether it has wider application or is unsafe to use.

12. Finalise report

Finalise documentation and report for the whole project. Make sure to include the preparatory study, investigate procedures and the data obtained from each activity. Comment on the project's success and viability.

Risks

This section covers the significant risks that could prevent this project from achieving its aim.

The main risk to this project is technical feasibility. It may not be possible to find a model that meets the desired design criteria and performance metrics.
To mitigate the above risk, careful review of design requirements and performance of previous models will be investigated. This will aim to solve over ambitious performance requirements.

Secondly, if a model is found it may not be possible for it to be practically implemented in a demonstration. These could be issues relating to scale where the position and direction of the robots cannot be tracked accurately enough for the avoidance system to be effective.

This will be mitigated by considering the demonstration throughout the model selection process. By understanding what kind of input the model will receive practically, the effects can be measured, and the performance of the demonstration can be predicted.

An important risk in any project is exceeding the allotted time duration. This risk will be mitigated by adhering to a strict project plan laid out before work commences. Slack will be added to all design-based activities that are likely to have unexpected work or delays. As a last resort, project scope can be changed if all the ECSA graduate attributes are adequately demonstrated in the work. This is an undesirable outcome but must be noted.

Another important risk is exceeding the budget. This is likely to impede the demonstration aspect of the project as all software required to create the model is provided under a student licence provided by the university. The main two components likely to cause this issue are the tracking system and the robots themselves. To mitigate this, a budget has been created and will be adhered to. Parts and components will be purchased well in advance of when they are required to avoid unexpected price changes and stock shortages that could result in more expensive components being purchased.

ECSA Outcome Self-assessment

A list of the ECSA graduate attributes and how this project will meet these requirements follows:

ECSA Graduate Attribute	Self-Assessment
1. Problem solving	This project provides a complex engineering problem that could have multiple solutions. Through the plan laid out in this proposal, a solution will be fully investigated and tested to see if it solves the stated problem effectively.
2. Application of scientific and engineering knowledge	This problem will require the use of many areas of knowledge. It contains using the basics of modelling and control systems, as well as combining computer systems and electrical design. All areas will need to be combined to create an effective solution.
3. Engineering design	This project will require creative problem solving to create an effective model as well as procedural design to test the effectiveness of the model's performance through data analysis. Non-procedural design and synthesis of components will be incorporated into the wheel-based robots in the demonstration.
4. Engineering methods, skills and tools, including information technology	Use of tools and skills will be critical in building the demonstration aspect of this project. These include practical skills such as 3D design and soldering to using data analysis techniques.
5. Professional and technical communication	This project will contain writing up a report of the whole investigation to be evaluated. This report shall be of a standard that can be read and understood by 4th year undergraduate engineers. The project will also be interrogated in an oral presentation.
6. Individual, team and multi-disciplinary working	This project consists of individual work, demonstrating the ability to work and solve problems without reliance on peers.
7. Independent learning	This project makes use of knowledge that has not been taught in lectures at SU, however the learning tools taught will be used to gain the required knowledge.

Table 1 ECSA Graduate Attribute Self-Assessment

Responsible Use of Resources and End-of-Life Strategies

The two outputs of this are a simulation model and a demonstration of the collision avoidance system.

The simulation product makes use of one main resource compute. This is expected to be done on a private laptop, as major computing capability is not required. This reduces the effect on the environment compared to a workstation PC and the compute demands are unlikely to be high. Network resources need to be considered for storage of data. Whilst the project is being worked on, it will be backed up both to Git, to a university OneDrive and have a local copy on the private laptop. The files for this project are unlikely to exceed 10GB so no significant strain will be put on any of these resources.

The simulation model can be provided to future students who are doing a follow-on research project that may require them to build a vehicle that requires a collision avoidance system. The simulation will be backed up to a location that the supervisor can advise on. It will need to be a location that can be accessed when required and is will not be over written or destroyed like a student OneDrive account.

The demonstration component will make use of a variety of materials. Firstly, electronic components for the tracking system and robots and likely 3D prints, wood and other construction materials. If possible, these will be reused off projects in the facility, however if unavailable, will be purchased. The parts made by construction materials will be made from recycled materials or reused materials wherever possible.

The demonstration component has multiple options for its end of life. Firstly, if successful, it could serve as a good demonstration for high school students attending open day or winter week activities at the university. It would be a hands-on and interactive tool for teaching engineering. If that is deemed to not be the desired outcome, the demonstration would be disassembled. The electronic parts and components such as the tracking technology and the microcontrollers from the robots would be set aside for use in other masters and final-year projects. The other materials that will be used will be recycled responsibly.

Conclusion

This project aims to create a universal proactive cooperative collision avoidance system that can be implemented cheaper than current market offerings. This proposed project, will investigate the current models and identify the best performing model for this project. It will then be created into a multibody simulation and demonstrated practically.

The project team has all the expertise at its disposal to successfully complete this project. It is expected that the total costs will be R297000, of which R5000 will be capital expenditure. The project is expected to take 8 months to complete. Mitigation and reduction measures have been taken to address the risks outlined in this proposal.

References

Mtimkulu, T. 2026. "Cooperative Collision Avoidance for the Voyager Unmanned Ground Vehicle: Practical Implementation and Demonstration". Electronic Engineering Master's Thesis. Department of Electrical and Electronic Engineering. Stellenbosch University.

Hughes, M. 2024. "Autonomous Guidance and Conflict Avoidance for Multiple Unmanned Aerial Vehicles (UAVs) in Urban Environments". Electronic Engineering Master's Thesis. Department of Electrical and Electronic Engineering. Stellenbosch University.

Appendix A: Budget

	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW/EEW			**TOTAL**
						Labour		Material	
	hr	R	R	R	R	hr	R	R	
Perform a literature and market review	25	12500							12500
Identify compatible applications	5	2500							2500
Compile design and performance criteria	5	2500							2500
Select feasible models	5	2500							2500
Simulate feasible models to determin performance	100	50000							50000
Select the highest performing model	8	4000							4000
Design Review	3	1500							1500
Create a multi-body performance simulation	80	40000							40000
Establish demonstration requirements and limitations	5	2500							2500
Create a scale demonstration of the best model	140	70000	1000		5000	80	28000	10000	114000
Validate simulation with demonstration	30	15000							15000
Finalise report	100	50000							50000
TOTAL	506	253000	1000	0	5000	80	28000	10000	297000

Table 2 Estimated Budget

The line item for the creating the scale demonstration includes the cost of the tracking technology (likely to be QR Code tracking, but others are being considered) and the parts for creating the wheel-based robots. It should be achievable within budget however if it cannot be achieved, a discussion can be had with the E&E faculty about getting hold of the required technology outside of the allocated budget.

Appendix A: Gantt Chart

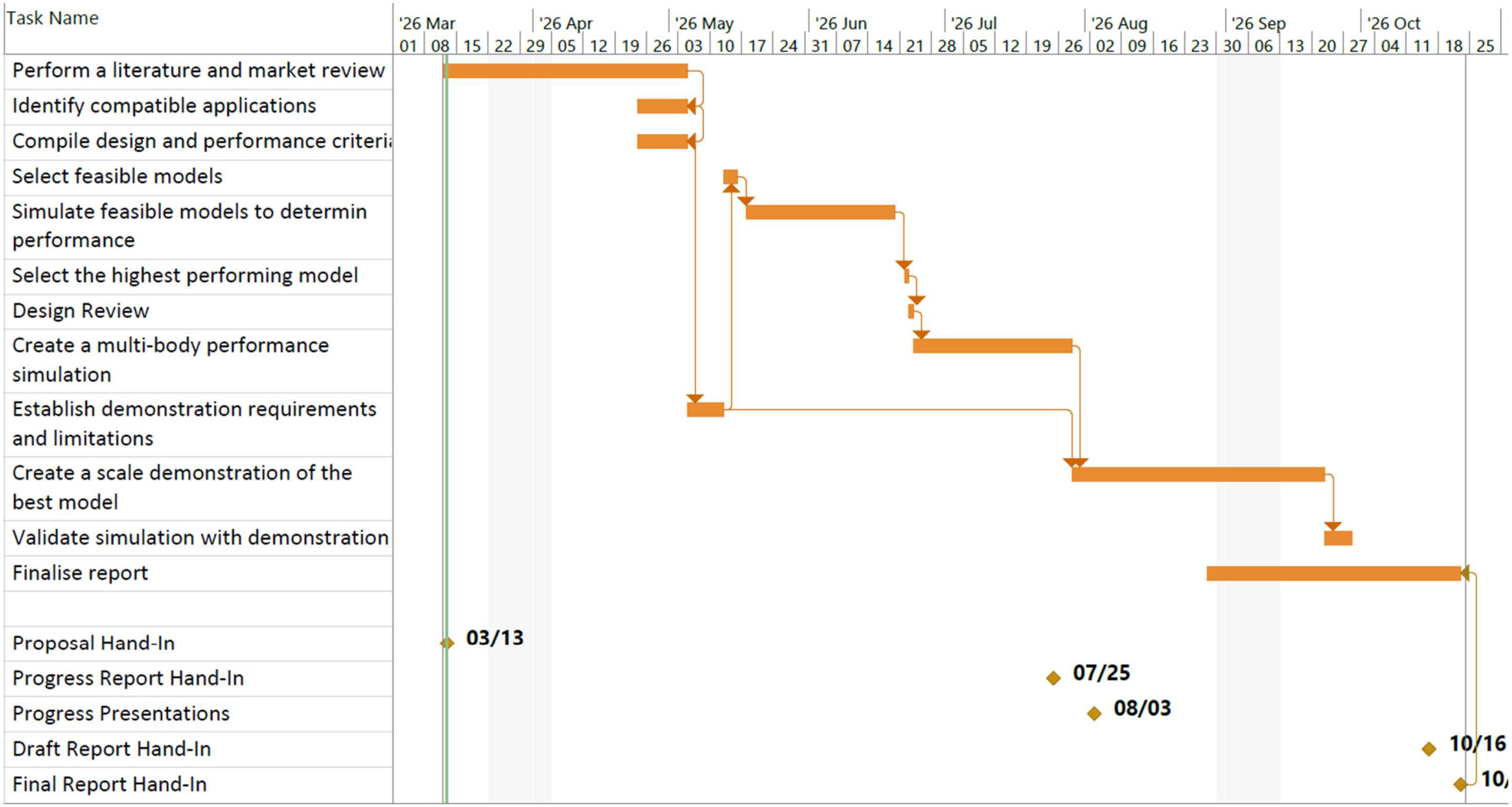


Figure 1 Preliminary Gantt Chart of Planned Activities